

LANG SON, Viet Nam

WMO: 488300

Lat: 21.8308N

Lon: 106.7690E

Elev: 863

StdP: 14.24

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	41.2	43.9	30.8	25.7	52.1	34.4	30.0	50.9	19.9	51.0	18.1	51.8	5.3	0	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	93.2	79.5	91.6	79.2	90.1	78.8	81.4	89.4	80.7	88.5	80.0	87.2	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	156.7	85.1	78.6	153.5	84.5	78.1	150.6	83.8	46.1	90.0	45.2	88.8	44.5	87.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.3	13.8	11.6	DB	36.7	96.7	2.8	1.6	34.7	97.9	33.0	98.8	31.4	99.7	29.3	100.9	
			WB	36.0	83.2	2.8	0.8	33.9	83.8	32.3	84.2	30.7	84.6	28.7	85.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	56.8	60.5	65.7	73.6	79.0	81.9	82.0	81.3	79.0	74.3	67.2	59.4
	DBStd	10.60	7.70	8.80	7.42	6.21	4.49	2.86	2.68	2.61	3.40	4.39	5.85	6.47
	HDD50	35	21	9	1	0	0	0	0	0	0	0	0	4
	HDD65	795	274	179	85	11	0	0	0	0	0	2	43	201
	CDD50	7981	232	303	488	708	899	956	992	969	870	753	515	296
	CDD65	3265	20	53	107	269	434	506	527	504	420	290	108	27
	CDH74	27047	50	219	414	1715	3829	5170	5311	4591	3334	1782	545	87
	CDH80	8853	2	40	91	463	1331	1878	1906	1531	1066	457	83	5
(14) Wind	WSAvg	4.0	5.2	4.9	4.6	4.3	3.8	3.0	2.9	2.8	3.4	3.9	4.4	5.0
(15) Precipitation	PrecAvg	51.8	1.4	1.3	2.2	3.0	6.4	7.6	10.0	8.5	6.0	2.5	1.8	1.1
	PrecMax	67.3	5.3	8.9	6.9	6.6	11.6	12.8	21.0	14.6	15.3	6.5	6.4	3.1
	PrecMin	34.8	0.0	0.0	0.3	0.6	2.7	3.8	5.2	3.1	0.9	0.4	0.0	0.0
	PrecStd	8.7	1.3	1.7	1.7	1.5	2.2	2.4	3.9	3.1	3.1	1.9	1.7	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.0	84.6	87.9	92.9	95.2	94.8	94.8	93.7	91.8	88.7	85.3	80.2
		MCWB	67.8	69.7	70.2	76.3	79.9	80.5	80.4	79.8	78.5	75.0	73.3	70.5
	2%	DB	74.9	80.2	82.3	88.3	91.9	92.3	92.5	91.8	89.8	86.6	82.4	76.5
		MCWB	65.8	69.5	72.4	76.5	78.9	80.1	79.8	79.3	77.7	74.3	72.0	67.7
	5%	DB	71.6	76.3	78.3	85.3	89.4	90.5	90.8	89.9	88.2	84.7	79.8	73.2
		MCWB	64.4	68.0	71.2	75.6	78.0	79.6	79.4	79.0	77.2	73.5	70.1	65.3
	10%	DB	68.5	72.9	75.2	82.4	87.0	88.6	88.8	87.8	86.2	82.6	77.1	69.9
		MCWB	63.4	66.8	70.0	74.2	77.2	79.3	79.1	78.6	76.5	72.9	68.4	63.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.7	71.9	74.8	78.8	81.5	82.4	82.3	81.8	80.8	78.0	75.6	72.3
		MCDB	75.4	79.4	82.2	87.7	90.5	90.8	90.9	89.6	88.1	84.9	81.6	78.0
	2%	WB	67.5	70.5	73.5	77.4	80.2	81.3	81.0	80.8	79.6	76.4	74.1	69.7
		MCDB	72.5	77.1	79.6	85.4	88.8	89.2	89.1	88.4	86.4	82.8	79.2	74.5
	5%	WB	66.0	69.2	72.3	76.3	79.2	80.5	80.2	79.9	78.7	75.5	72.7	67.5
		MCDB	70.2	74.8	77.1	83.4	87.0	88.0	87.8	87.0	85.0	81.5	77.3	71.2
	10%	WB	64.3	67.8	71.0	75.2	78.3	79.8	79.6	79.1	77.9	74.6	71.2	65.0
		MCDB	68.1	72.1	74.9	81.3	85.1	86.6	86.4	85.5	83.5	80.0	75.6	68.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.5	10.0	11.3	12.5	12.3	12.0	11.8	12.6	13.8	14.7	13.6
		MCDBR	17.3	17.3	15.1	15.6	16.3	15.4	15.7	15.7	15.7	17.1	18.4	17.8
	5% WB	MCWBR	8.5	7.6	6.4	5.8	5.4	5.2	4.9	5.0	5.0	6.1	7.9	8.5
		MCWBR	13.4	14.7	12.2	14.0	13.8	13.4	13.8	13.9	13.4	13.9	14.4	13.7
(40) Clear-Sky Solar Irradiance	taub		0.578	0.609	0.784	0.741	0.550	0.527	0.499	0.562	0.601	0.690	0.575	0.599
		taud	1.758	1.733	1.446	1.564	2.038	2.120	2.187	1.986	1.851	1.623	1.817	1.731
	Ebn at Noon		217	221	190	203	244	248	255	240	228	198	216	204
		Edh at Noon	66	71	99	89	55	50	47	58	65	79	62	66
(44) All-Sky Solar Radiation	RadAvg		671	793	861	1117	1387	1445	1474	1441	1394	1197	1009	790
	RadStd		164	143	115	127	141	144	122	101	92	90	145	140

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.72	+0.54	+0.49	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	+1.08	+0.36	+0.34	N/A	N/A	N/A	+144

Nomenclature: See separate page